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Semiconductor device, semiconductor module, and wireless communication apparatus

Abstract

A semiconductor device including: a channel layer; a barrier layer; a source electrode and a drain electrode; a gate electrode; a side surface opening region; and a low-Ns region. The channel layer includes a first nitride semiconductor. The barrier layer includes a second nitride semiconductor. The barrier layer is provided on the channel layer. The source electrode and the drain electrode are provided above the barrier layer. The gate electrode is provided above the barrier layer between the source electrode and the drain electrode. The side surface opening region is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode. The low-Ns region is provided in the channel layer in correspondence with a planar region provided with the gate electrode and the side surface opening region. The low-Ns region has lower carrier density than carrier density of another region of the channel layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2021/023166 filed on Jun. 18, 2021, which claims priority benefit of Japanese Patent Application No. JP 2020-123446 filed in the Japan Patent Office on Jul. 20, 2020. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a semiconductor device, a semiconductor module, and a wireless communication apparatus.

BACKGROUND ART

(3) In recent years, high electron mobility transistors (High Electron Mobility Transistors: HEMTs) have been proposed that each use a two-dimensional electron gas layer formed at a heterojunction interface between compound semiconductors as a channel.

(4) A two-dimensional electron gas layer has high electron mobility and high sheet electron density. HEMT that uses the two-dimensional electron gas layer as a channel is promising as a low-resistance and high-withstand-voltage transistor that is able to perform a high-speed operation. For example, the HEMT is expected to be applied to a power device, an RF (Radio Frequency) device, or the like.

(5) For example, HEMT having a MIS (Metal-Insulator-Semiconductor) structure has been proposed in which a compound semiconductor layer and a gate metal are bonded with a gate insulating film interposed in between (e.g., PTL 1). The HEMT having a MIS structure includes the gate insulating film at the interface between the compound semiconductor layer and the gate and gate leak is suppressed during an off operation. This makes it possible to reduce power consumption. Therefore, the HEMT having a MIS structure is expected to be suitable for a portable wireless communication apparatus.

CITATION LIST

Patent Literature

(6) PTL 1: Japanese Unexamined Patent Application Publication No. 2017-92083

SUMMARY OF THE INVENTION

(7) In general, thinning down the gate insulating film allows the transistor having a MIS structure to achieve a higher-speed operation. Meanwhile, thinning down the gate insulating film causes the dielectric voltage of the gate insulating film to decrease. This may decrease the gate withstand voltage during an off operation.

(8) In the MIS-type HEMT, the two-dimensional electron gas layer serving as a channel has high carrier density. The gate insulating film thus receives a higher electric field more easily. The HEMT is therefore requested to have a higher gate withstand voltage.

(9) It is desirable to provide a semiconductor device having higher withstand voltage performance and a semiconductor module and a wireless communication apparatus each including the semiconductor device.

(10) A semiconductor device according to an embodiment of the present disclosure includes: a channel layer; a barrier layer; a source electrode and a drain electrode; a gate electrode; a side surface opening region; and a low-Ns region. The channel layer includes a first nitride semiconductor. The barrier layer includes a second nitride semiconductor. The barrier layer is provided on the channel layer. The source electrode and the drain electrode are provided above the barrier layer. The gate electrode is provided above the barrier layer between the source electrode

and the drain electrode. The side surface opening region is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode. The low-Ns region is provided in the channel layer in correspondence with a planar region provided with the gate electrode and the side surface opening region. The low-Ns region has lower carrier density than carrier density of another region of the channel layer.

(11) A semiconductor module according to an embodiment of the present disclosure includes a semiconductor device. The semiconductor device includes a channel layer, a barrier layer, a source electrode and a drain electrode, a gate electrode, a side surface opening region, and a low-Ns region. The channel layer includes a first nitride semiconductor. The barrier layer includes a second nitride semiconductor. The barrier layer is provided on the channel layer. The source electrode and the drain electrode are provided above the barrier layer. The gate electrode is provided above the barrier layer between the source electrode and the drain electrode. The side surface opening region is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode. The low-Ns region is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region. The low-Ns region has lower carrier density than carrier density of another region of the channel layer.

(12) A wireless communication apparatus according to an embodiment of the present disclosure includes a semiconductor device. The semiconductor device includes a channel layer, a barrier layer, a source electrode and a drain electrode, a gate electrode, a side surface opening region, and a low-Ns region. The channel layer includes a first nitride semiconductor. The barrier layer includes a second nitride semiconductor. The barrier layer is provided on the channel layer. The source electrode and the drain electrode are provided above the barrier layer. The gate electrode is provided above the barrier layer between the source electrode and the drain electrode. The side surface opening region is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode. The low-Ns region is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region. The low-Ns region has lower carrier density than carrier density of another region of the channel layer.

(13) In the semiconductor device, the semiconductor module, and the wireless communication apparatus according to the respective embodiments of the present disclosure, it is possible to control the polarization or the like of the barrier layer exposed by the gate opening provided with the gate electrode and the side surface opening region at least provided on one of the side surfaces of the gate electrode. This makes it possible to form the low-Ns region having reduced carrier density in the channel layer corresponding to the barrier layer.

Description

BRIEF DESCRIPTION OF DRAWING

(1) FIG. 1 is a vertical cross-sectional view of a configuration of a semiconductor device according to an embodiment of the present disclosure.

(2) FIG. 2 is a top view of the configuration of the semiconductor device according to the embodiment.

(3) FIG. 3 is an enlarged vertical cross-sectional view of a region near a gate electrode and a side surface opening region in FIG. 1.

(4) FIG. 4 is a graphical chart illustrating a relationship between carrier density of a low-Ns region and a withstand voltage of the semiconductor device.

(5) FIG. 5A is a vertical cross-sectional view describing a step of a method of manufacturing the semiconductor device according to the embodiment.

- (6) FIG. 5B is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (7) FIG. 5C is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (8) FIG. 5D is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (9) FIG. 5E is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (10) FIG. 5F is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (11) FIG. 5G is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device according to the embodiment.
- (12) FIG. 6 is a vertical cross-sectional view describing a modification example of a method of forming the low-Ns region.
- (13) FIG. 7A is an energy band diagram of the semiconductor device in which fixed electric charge has not yet been applied to a barrier layer.
- (14) FIG. 7B is an energy band diagram of the semiconductor device in which the fixed electric charge has already been applied to the barrier layer.
- (15) FIG. 8 is a schematic perspective view of a configuration of a semiconductor module according to a second embodiment of the present disclosure.
- (16) FIG. 9 is a block diagram illustrating a configuration of a wireless communication apparatus according to a third embodiment of the present disclosure.

MODES FOR CARRYING OUT THE INVENTION

- (17) The following describes embodiments of the present disclosure in detail with reference to the drawings. The embodiments described below are specific examples of the present disclosure. The technology according to the present disclosure is not limited to the following modes. In addition, the disposition, dimensions, dimensional ratios, and the like of the respective components according to the present disclosure are not limited to the modes illustrated in the drawings.
- (18) It is to be noted that description is given in the following order. 1. First Embodiment 1.1. Configuration Example 1.2. Manufacturing Method 2. Second Embodiment 3. Third Embodiment
- #### 1. FIRST EMBODIMENT
- (19) (1.1. Configuration Example)
- (20) First, a configuration of a semiconductor device according to a first embodiment of the present disclosure is described with reference to FIGS. 1 and 2. FIG. 1 is a vertical cross-sectional view of a configuration of a semiconductor device **100** according to the present embodiment. FIG. 2 is a top view of the configuration of the semiconductor device **100** according to the present embodiment.
- (21) As illustrated in FIG. 1, the semiconductor device **100** includes a substrate **110**, a buffer layer **120**, a channel layer **130**, a low-Ns region **131**, a barrier layer **140**, an ohmic layer **151**, a source electrode **150S**, a drain electrode **150D**, a first insulating layer **161**, a second insulating layer **162**, a third insulating layer **163**, a gate electrode **170**, a gate insulating film **171**, and a side surface opening region **172**.
- (22) In other words, the semiconductor device **100** is a high electron mobility transistor (HEMT) that uses a two-dimensional electron gas layer (2DEG) as a channel. The two-dimensional electron gas layer (2DEG) is formed by using the difference between the magnitude of the polarization of the channel layer **130** and the magnitude of the polarization of the barrier layer **140**. The two-dimensional electron gas layer is formed, for example, in the channel layer **130** near the barrier layer **140**.
- (23) The substrate **110** is a supporting body of the semiconductor device **100**. Specifically, the substrate **110** may be a substrate including a semiconductor material having a lattice constant close

to that of the nitride semiconductor included in the channel layer **130**. For example, the substrate **110** may be a substrate including a III-V group compound semiconductor such as a single-crystal GaN substrate.

(24) It is to be noted that the semiconductor device **100** is provided with the buffer layer **120** and it is thus also possible to use, as the substrate **110**, a substrate including a material having a lattice constant different from that of the channel layer **130**. For example, the substrate **110** may be a SiC substrate, a sapphire substrate, a Si substrate, or the like.

(25) The buffer layer **120** is a nitride semiconductor layer epitaxially grown on the substrate **110**. Control over the lattice constant of the surface provided with the channel layer **130** allows the buffer layer **120** to reduce the lattice mismatch between the substrate **110** and the channel layer **130**. This allows the buffer layer **120** to offer a more favorable crystal state for the channel layer **130** and suppress the warpage of the substrate **110**. For example, in a case where the substrate **110** is a single-crystal Si substrate and the channel layer **130** is a GaN layer, the buffer layer **120** may include AlN, AlGa_N, or GaN. However, depending on the configurations of the substrate **110** and the channel layer **130**, the buffer layer **120** may be omitted.

(26) The buffer layer **120** may be provided to have a monolayer structure. Alternatively, the buffer layer **120** may be provided to have a multilayer structure in which a plurality of layers of AlN, AlGa_N, or GaN is stacked. In addition, in a case where the buffer layer **120** includes a ternary material such as AlGa_N, the buffer layer **120** may be provided to gradually change in composition in the thickness direction.

(27) The channel layer **130** includes a nitride semiconductor having a narrower band gap than that of the barrier layer **140**. The channel layer **130** allows carriers to be accumulated on the channel layer **130** side near the interface between the channel layer **130** and the barrier layer **140** by using the difference from the magnitude of the polarization of the barrier layer **140**. For example, the channel layer **130** may include epitaxially grown GaN. In addition, the channel layer **130** may include undoped u-GaN to which no impurity is added. In such a case, the channel layer **130** makes it possible to suppress the impurity scattering of the carriers. It is thus possible to increase the carrier mobility.

(28) The barrier layer **140** includes a nitride semiconductor having a wider band gap than that of the channel layer **130**. The barrier layer **140** bonded to the channel layer **130** allows carriers to be accumulated in the channel layer **130** near the barrier layer **140** by spontaneous polarization or piezoelectric polarization. This forms a two-dimensional electron gas layer (2DEG) having high mobility and high carrier concentration in the channel layer **130** near the barrier layer **140**.

(29) For example, the barrier layer **140** may include epitaxially grown Al_{sub.1-x}Ga_{sub.x}In_{sub.y}N (where $0 \leq x < 1$ and $0 \leq y < 1$). In addition, the barrier layer **140** may include undoped u-Al_{sub.1-x-y}Ga_{sub.x}In_{sub.y}N to which no impurity is added. In such a case, the barrier layer **140** makes it possible to suppress the impurity scattering of the carriers in the channel layer **130**. It is possible to further increase the carrier mobility.

(30) It is to be noted that the barrier layer **140** may include a single layer. Alternatively, the barrier layer **140** may include a plurality of stacked layers of Al_{sub.1-x-y}Ga_{sub.x}In_{sub.y}N that is different in composition. In addition, the barrier layer **140** may be configured to gradually change in composition in the thickness direction.

(31) The low-Ns region **131** is a region that is provided in a partial region of the channel layer **130** and has lower carrier density than that of the other regions of the channel layer **130**. Specifically, the low-Ns region **131** is provided in the partial region of the channel layer **130** corresponding to the planar region provided with the gate electrode **170** and the side surface opening regions **172**. The low-Ns region **131** functions as a portion of the current path between the drain electrode **150D** and the source electrode **150S**.

(32) The carrier density of the low-Ns region **131** is reduced as compared with that of the other regions of the channel layer **130**. This makes it possible to suppress the application of an excessive

electric field to the gate insulating film **171** between the low-Ns region **131** and the gate electrode **170**. In addition, the low-Ns region **131** is provided to be wider from the gate electrode **170** by the side surface opening regions **172**. This makes it possible to suppress the occurrence of the concentration of an electric field at the end of the gate electrode **170**. This allows the low-Ns region **131** to increase the withstand voltage performance of the semiconductor device **100**.

(33) In addition, as illustrated in FIG. 2, there is provided an active region **140A** in the stacked structure of the substrate **110**, the buffer layer **120**, the channel layer **130**, and the barrier layer **140**. The active region **140A** is surrounded by an insulative element isolation region.

(34) The active region **140A** is a region provided with the respective components of the semiconductor device **100**. The active region **140A** is provided as an island-shaped region surrounded by the element isolation region. The element isolation region is provided by inactivating the channel layer **130** and the barrier layer **140** by implanting ions such as B (boron). The element isolation region is provided to surround the active region **140A** and electrically isolates the active regions **140A** from each other. It is to be noted that the element isolation region may be formed by removing the channel layer **130** and the barrier layer **140** by etching.

(35) The active region **140A** is provided with the source electrode **150S** and the drain electrode **150D** in the extending direction of the active region **140A**. The active region **140A** is provided with the gate electrode **170** between the source electrode **150S** and the drain electrode **150D**. Specifically, the source electrode **150S** and the drain electrode **150D** may be provided to protrude to the opposite sides in the direction orthogonal to the extending direction of the active region **140A**. The gate electrode **170** may be provided to cross the active region **140A** in the direction orthogonal to the extending direction of the active region **140A**.

(36) The ohmic layers **151** are provided on both sides of the gate electrode **170** on the barrier layer **140**. Each of the ohmic layers **151** forms an ohmic contact with the barrier layer **140** through heat treatment. This makes it possible to reduce the contact resistance between the source electrode **150S** and the drain electrode **150D** provided in the upper layer, and the two-dimensional electron gas layer (2DEG). The ohmic layer **151** may be provided to have a structure in which, for example, Ti (titanium), Al (aluminum), Ni (nickel), and Au (gold) are sequentially stacked from the barrier layer **140** side.

(37) The source electrode **150S** and the drain electrode **150D** are provided on the ohmic layers **151** provided on both sides of the gate electrode **170**. It is possible to electrically couple the source electrode **150S** and the drain electrode **150D** to the two-dimensional electron gas layer (2DEG) in the channel layer **130** through the ohmic layers **151**. The source electrode **150S** and the drain electrode **150D** may be each provided to have a structure in which Ti (titanium), Al (aluminum), Ni (nickel), and Au (gold) are sequentially stacked from the ohmic layer **151** side.

(38) It is to be noted that there may be provided a regrowth layer below each of the ohmic layers **151**. The regrowth layer is provided, for example, by digging the barrier layer **140** and the channel layer **130** from the barrier layer **140** side to a region deeper than the interface between the barrier layer **140** and the channel layer **130** and filling the dug region with an n-type nitride semiconductor.

(39) Specifically, the regrowth layer may be provided selectively growing a nitride semiconductor including an n-type impurity epitaxially in each of the regions corresponding to the source and the drain by using a selection mask. Alternatively, the regrowth layer may be provided epitaxially growing a nitride semiconductor including an n-type impurity on the entire surface and then removing the deposit on the mask.

(40) Each of the regrowth layers has higher electrical conductivity than that of the barrier layer **140**. This makes it possible to electrically couple the source electrode **150S** and the drain electrode **150D**, and the two-dimensional electron gas layer (2DEG) with lower resistance. For example, the regrowth layer may include GaN including an n-type impurity such as Si or Ge at $1.0 \times 10^{19} / \text{cm}^3$ or more. Alternatively, the regrowth layer may include $\text{In}_{1-x}\text{Ga}_x\text{N}$ ($0 \leq x < 1$), which is easier than GaN to grow at a low temperature.

(41) The first insulating layer **161** is provided by using an insulative material to cover the barrier layer **140** and the ohmic layers **151**. Specifically, the first insulating layer **161** is provided by using a material having an insulation property with respect to the barrier layer **140**, the ohmic layers **151**, and the gate electrode **170**. This allows the first insulating layer **161** to protect the surface of the barrier layer **140** from an impurity such as ions and suppress a decrease in the characteristics of the semiconductor device **100** by offering a favorable interface with the barrier layer **140**. For example, the first insulating layer **161** may include a film of Al.sub.2O.sub.3 that has a film thickness of about 50 nm.

(42) In addition, the first insulating layer **161** is provided with openings **150H** for providing the source electrode **150S** and the drain electrode **150D** and a gate opening **170H** for providing the gate electrode **170**. To allow the side surface opening regions **172** described below to be provided, the gate opening **170H** provided in the first insulating layer **161** is provided to have a greater width than that of the gate opening **170H** provided in the second insulating layer **162**.

(43) The second insulating layer **162** is provided by using an insulative material to cover the first insulating layer **161**. The second insulating layer **162** is provided by using a material having an insulation property with respect to the first insulating layer **161**, the gate electrode **170**, and the gate insulating film **171**. This allows the second insulating layer **162** to protect the surface of the barrier layer **140** from an impurity such as ions. For example, the second insulating layer **162** may include a film of SiO.sub.2 that has a film thickness of about 100 nm.

(44) In addition, the second insulating layer **162** is provided with the openings **150H** for providing the source electrode **150S** and the drain electrode **150D** and the gate opening **170H** for providing the gate electrode **170**. The gate opening **170H** provided in the second insulating layer **162** has a smaller width than that of the gate opening **170H** provided in the first insulating layer **161**. The second insulating layer **162** is therefore provided to protrude from the first insulating layer **161** inside the gate opening **170H**.

(45) The gate insulating film **171** is uniformly provided by using an insulative material on the second insulating layer **162** along the shapes of the openings **150H** and the gate opening **170H**. Specifically, the gate insulating film **171** is uniformly provided on the top of the second insulating layer **162**, the inner side surfaces of the openings **150H** provided in the first insulating layer **161** and the second insulating layer **162**, and the inner side surfaces and the inner bottom surface of the gate opening **170H**. The gate insulating film **171** is provided by using a material having an insulation property with respect to the barrier layer **140** and the gate electrode **170**. This allows the gate insulating film **171** to protect the surface of the barrier layer **140** from an impurity such as ions and suppress a decrease in the characteristics of the semiconductor device **100** by offering a favorable interface with the barrier layer **140**. For example, the gate insulating film **171** may be provided as a single layer film or a stacked film of Al.sub.2O.sub.3 or HfO.sub.2 that has a film thickness of about 10 nm.

(46) The gate electrode **170** is provided by using an electrically conductive material on the gate insulating film **171** inside the gate opening **170H**. Specifically, the gate electrode **170** is provided on the gate insulating film **171** on the barrier layer **140** from the top of the second insulating layer **162** through the gate opening **170H**. For example, the gate electrode **170** may be provided by stacking Ni (nickel) and Au (gold) from the gate insulating film **171** side.

(47) The gate electrode **170** makes it possible to control the carrier concentration of the two-dimensional electron gas layer (2DEG) formed in the channel layer **130** by using an applied voltage. Specifically, the gate electrode **170** makes it possible to control the carrier concentration of the two-dimensional electron gas layer (2DEG) formed in the channel layer **130** with an electric field effect by controlling the thickness of a depletion layer formed in the barrier layer **140** below with an application voltage.

(48) The side surface opening region **172** is at least provided on one of the side surfaces of the gate electrode **170** between the source electrode **150S** or the drain electrode **150D** and the gate electrode

170 as a side surface opening for the first insulating layer **161**. The side surface opening region **172** allows the low-Ns region **131** to be formed in the channel layer **130** by processing the barrier layer **140** through the side surface opening region **172** and the gate opening **170H**. In other words, the low-Ns region **131** is provided in the channel layer **130** in the planar region provided with the side surface opening region **172** and the gate electrode **170**.

(49) For example, the low-Ns region **131** may be formed by recessing the barrier layer **140** exposed by the side surface opening region **172** and the gate opening **170H**, adding fixed electric charge to the exposed barrier layer **140**, or doping the exposed barrier layer **140** with anions.

(50) The side surface opening region **172** may be at least provided on the side surface of the gate electrode **170** between the drain electrode **150D** and the gate electrode **170**. The concentration of an electric field at the end of the gate electrode **170** is likely to occur on the drain electrode **150D** side. The formation of the low-Ns region **131** on the drain electrode **150D** side therefore allows the side surface opening region **172** to increase the withstand voltage performance at the end of the gate electrode **170**.

(51) In addition, the respective side surface opening regions **172** may be provided on both of the side surfaces of the gate electrode **170** between the source electrode **150S** and the drain electrode **150D**, and the gate electrode **170**. This allows the side surface opening regions **172** to further increase the withstand voltage performance at the end of the gate electrode **170** by forming the low-Ns regions **131** on the respective sides of the source electrode **150S** and the drain electrode **150D**.

(52) It is to be noted that, in such a case, the side surface opening region **172** provided on the drain electrode **150D** side may be provided to have a greater width than that of the side surface opening region **172** provided on the source electrode **150S** side. As described above, the concentration of an electric field at the end of the gate electrode **170** is likely to occur on the drain electrode **150D** side. Therefore, in a case where the side surface opening region **172** on the drain electrode **150D** side and the low-Ns region **131** are each provided to have a greater width, the semiconductor device **100** is allowed to have further higher withstand voltage performance.

(53) In addition, as illustrated in FIG. 2, the side surface opening region **172** may be provided over the entire circumference of the side surfaces of the gate electrode **170**. For example, the side surface opening region **172** is formed by forming the gate opening **170H** by dry etching and removing the side surfaces of the first insulating layer **161** exposed by the gate opening **170H** by wet etching. The gate opening **170H** extends through the first insulating layer **161** and the second insulating layer **162**. In such a case, the wet etching proceeds isotropically. This causes the side surface opening region **172** to be provided to extend from the gate opening **170H** on the entire circumference. The side surface opening region **172** allows the low-Ns region **131** to be formed in the active region **140A** immediately below the side surface opening region **172** and it is thus possible to suppress the application of an excessive electric field and the concentration of an electric field between the low-Ns region **131** and the gate electrode **170**.

(54) The third insulating layer **163** is provided by using an insulative material on the gate insulating film **171**. The third insulating layer **163** is provided by using a material having an insulation property with respect to the source electrode **150S**, the drain electrode **150D**, and the gate electrode **170**. For example, the third insulating layer **163** may include SiO₂, Si₃N₄, or the like. It is to be noted that the third insulating layer **163** is provided with the openings **150H** for providing the source electrode **150S** and the drain electrode **150D**.

(55) As described above, the semiconductor device **100** according to the present embodiment is provided with the low-Ns region **131** having reduced carrier density in the partial region of the channel layer **130**. This makes it possible to increase the gate withstand voltage of the semiconductor device **100**.

(56) It is possible to set as appropriate, by the specifications of the semiconductor device **100**, the size of the low-Ns region **131** and the degree to which the carrier density of the low-Ns region **131** is reduced. However, further enlarging the low-Ns region **131** and further reducing the carrier

density of the low-Ns region **131** may increase the on-resistance of the semiconductor device **100** while allowing the semiconductor device **100** to have higher withstand voltage performance.

(57) With reference to FIGS. **3** and **4**, the following specifically describes the favorable size of the low-Ns region **131** and the favorable degree to which the carrier density of the low-Ns region **131** is reduced, which allows the semiconductor device **100** to favorably achieve both the withstand voltage performance and the on-resistance.

(58) (Favorable Size of Low-Ns Region **131**)

(59) FIG. **3** is an enlarged vertical cross-sectional view of a region near the gate electrode **170** and the side surface opening regions **172** in FIG. **1**. In a case where a gate length w_6 of the gate electrode **170** is L_g , it is preferable that the side surface opening regions **172** each have a width w of $0.3 L_g$ or more and $0.4 L_g$ or less from the gate electrode **170** and the low-Ns region **131** have a width w_5 of $1.6 L_g$ or more and $1.8 L_g$ or less as illustrated in FIG. **3**. In particular, in a case where the barrier layer **140** includes AlInN having an In concentration of 19%, providing the side surface opening regions **172** and the low-Ns region **131** with the widths within the ranges described above makes it possible to prevent the on-resistance of the semiconductor device **100** from excessively increasing while increasing the withstand voltage performance.

(60) Alternatively, it is preferable that the side surface opening regions **172** each have a width w of 100 nm or less from the gate electrode **170** and it is preferable that the low-Ns region **131** be provided in a region that protrudes to both the source electrode **150S** side and the drain electrode **150D** side from the gate electrode **170** within a range of 100 nm or less. In particular, in a case where the barrier layer **140** includes AlInN having an In concentration of 19%, providing the side surface opening regions **172** with the widths within the range described above and forming the low-Ns region **131** within an appropriate range make it possible to prevent the on-resistance of the semiconductor device **100** from excessively increasing while increasing the withstand voltage performance. It is to be noted that the lower limit value of the width of each of the side surface opening regions **172** is not particularly limited, but the width of the side surface opening region **172** may have a lower limit value of 10 nm from the perspective of formation difficulty and cost.

(Favorable Degree to Which Carrier Density of Low-Ns Region **131** Is Reduced)

(61) As described above, it is possible to set as appropriate, in accordance with the specifications of the semiconductor device **100**, the degree to which the carrier density of the low-Ns region **131** is reduced. However, it is preferable that the carrier density of the low-Ns region **131** be 50% or more and 90% or less of the carrier density of the other regions of the channel layer **130**. In such a case, the semiconductor device **100** is allowed to have higher withstand voltage performance and have an excessive increase in on-resistance suppressed.

(62) FIG. **4** is a graphical chart illustrating a relationship between the carrier density (Ns) of the low-Ns region **131** and the withstand voltage of the semiconductor device **100**. FIG. **4** illustrates results of the measurement of the withstand voltages of the respective semiconductor devices **100** manufactured by fluctuating the carrier density (Ns) of the low-Ns region **131** by using values obtained by standardizing, as 1, the carrier density and the withstand voltage of the semiconductor device **100** including the low-Ns region **131** having the highest carrier density.

(63) It is to be noted that the withstand voltage is indicated as a gate voltage obtained in a case where a drain current I_d and a gate voltage V_g of the semiconductor device **100** are evaluated and the I_d waveform exceeds the threshold value. For example, the gate voltage obtained in a case where the drain current I_d exceeds 1×10^{-6} mA/mm with a drain voltage V_d fixed at 0.1 V and the gate voltage V_g scanned in the off direction is indicated as a withstand voltage.

(64) As illustrated in FIG. **4**, the withstand voltage of the semiconductor device **100** is increased by reducing the carrier density of the low-Ns region **131** by 10%. In addition, it is indicated that the withstand voltage of the semiconductor device **100** is increased up to about three times by reducing the carrier density of the low-Ns region **131** by 50%. As indicated by the graphical chart illustrated in FIG. **4**, the withstand voltage of the semiconductor device **100** linearly increases as the carrier

density of the low-Ns region **131** is reduced. Even as compared with the other regions of the channel layer **130**, it is thus considered that the low-Ns region **131** similarly allows the semiconductor device **100** to have a higher withstand voltage by reducing the carrier density from that of the other regions of the channel layer **130** by 10% or more and 50% or less.

(65) It is thus preferable that the low-Ns region **131** have carrier density corresponding to 50% or more and 90% or less of that of the other regions of the channel layer **130**. This allows the semiconductor device **100** to have higher withstand voltage performance and have an excessive increase in on-resistance suppressed. In particular, in a case where the barrier layer **140** includes AlInN having an In concentration of 19%, the carrier density of the low-Ns region **131** within the range described above allows the semiconductor device **100** to achieve both the withstand voltage and the on-resistance more favorably.

(66) (1.2. Manufacturing Method)

(67) Subsequently, a method of manufacturing the semiconductor device **100** according to the present embodiment is described with reference to FIGS. 5A, 5B, 5C, 5D, 5E, 5F, and 5G. Each of FIGS. 5A, 5B, 5C, 5D, 5E, 5F, and 5G is a vertical cross-sectional view describing a step of the method of manufacturing the semiconductor device **100** according to the present embodiment.

(68) First, as illustrated in FIG. 5A, for example, the buffer layer **120** is epitaxially grown on the substrate **110** including Si or the like and GaN is then epitaxially grown to form the channel layer **130**. Subsequently, AlGa_{0.3}N (Al_{0.3}Ga_{0.7}N mixed crystal) is epitaxially grown on the channel layer **130** to form the barrier layer **140**.

(69) After that, the unillustrated ohmic layers **151** are formed and the unillustrated element isolation region is formed.

(70) Specifically, the ohmic layers **151** are formed on the barrier layer **140** on both sides of a region in which the gate electrode **170** is formed in a subsequent step. Performing annealing treatment forms the ohmic layers **151** that are electrically coupled to the barrier layer **140** with low contact resistance. It is also possible to provide a regrowth layer below each of the ohmic layers **151**. The regrowth layer is obtained by selectively growing an n-type nitride semiconductor.

(71) In addition, B (boron) is ion-implanted into predetermined planar regions of the barrier layer **140** and the channel layer **130** to form an element isolation region in which the resistance of the barrier layer **140** and the channel layer **130** is increased. The element isolation region is formed, for example, to surround the island-shaped active region **140A** and electrically isolates the active regions **140A** from each other. The element isolation region may be formed at any timing such as a timing after the formation of the gate electrode **170**.

(72) Subsequently, as illustrated in FIG. 5B, Al₂O₃ (aluminum oxide) is deposited on the barrier layer **140** by using an ALD (Atomic Layer Deposition) method to form the first insulating layer **161**. Further, SiO₂ (silicon oxide) is deposited on the first insulating layer **161** by using a CVD (Chemical Vapor Deposition) method to form the second insulating layer **162**.

(73) Next, as illustrated in FIG. 5C, the second insulating layer **162** is etched to form the gate opening **170H** that exposes the first insulating layer **161**. In this case, the use of dry etching allows the gate opening **170H** to be formed with an increase in opening width suppressed more.

(74) It is to be noted that, in a case where the second insulating layer **162** is etched to excessively etch the first insulating layer **161**, the barrier layer **140** may be damaged in a subsequent process. In a case where the etching selectivity between the first insulating layer **161** and the second insulating layer **162** is 1 or more, preferably 5 or more, the excessive etching of the first insulating layer **161** is suppressed. This makes it possible to suppress damage to the barrier layer **140**.

(75) Subsequently, as illustrated in FIG. 5D, the first insulating layer **161** is subjected to wet etching to form a side surface opening **172H** on the side surfaces of the first insulating layer **161** exposed by the gate opening **170H**. In a case where the etching selectivity between the first insulating layer **161** and the second insulating layer **162** is then 1 or more, preferably 5 or more, it is possible to suppress the procession of etching in the second insulating layer **162** while causing

etching to proceed in the side surface direction of the first insulating layer **161**. This makes it possible to suppress the wet etching of the first insulating layer **161** increasing the opening width of the gate opening **170H** formed in the second insulating layer **162**.

(76) Next, as illustrated in FIG. 5E, the barrier layer **140** exposed by the gate opening **170H** and the side surface opening **172H** is recessed (retracted) to form the low-Ns region **131** in the channel layer **130**. Specifically, it is possible to weaken the self-polarization (a mixture of self-polarization and piezoelectric polarization depending on a material) of the barrier layer **140** by recessing the barrier layer **140**. This makes it possible to reduce the density of carriers accumulated in the channel layer **130** in accordance with the difference between the magnitude of the polarization of the channel layer **130** and the magnitude of the polarization of the barrier layer **140**. This makes it possible to form the low-Ns region **131** in the channel layer **130** in the region obtained by recessing the barrier layer **140**. It is to be noted that it is possible to recess the barrier layer **140**, for example, by oxidating the barrier layer **140** and performing wet etching on the barrier layer **140**.

(77) In the method described above, it is possible to form the low-Ns region **131** with no mask or the like added because the low-Ns region **131** is formed in the channel layer **130** by using the gate opening **170H** and the side surface opening **172H**. It is to be noted that an end of the low-Ns region **131** may align with an end of the side surface opening **172H**. Alternatively, an end of the low-Ns region **131** may extend outward from an end of the side surface opening **172H**.

(78) Subsequently, as illustrated in FIG. 5F, Al.sub.2O.sub.3 (aluminum oxide) is deposited by using an ALD (Atomic Layer Deposition) method to form the gate insulating film **171**. The ALD method makes it possible to form a uniform film on all the exposed regions. The gate insulating film **171** is thus formed as a uniform film on the exposed surfaces of the barrier layer **140**, the first insulating layer **161**, and the second insulating layer **162** including the inside of the gate opening **170H** and the side surface opening **172H**.

(79) Next, as illustrated in FIG. 5G, Ni (nickel) and Au (gold) are sequentially deposited on the gate insulating film **171** to fill the gate opening **170H**. This forms the gate electrode **170**. After that, the third insulating layer **163** is deposited on the gate insulating film **171** and the source electrode **150S** and the drain electrode **150D** are then formed on the unillustrated ohmic layers **151** to make it possible to manufacture the semiconductor device **100** according to the present embodiment.

(80) It is to be noted that the side surface opening regions **172** are gaps in the manufacturing method described above, but it is also possible to fill each of the side surface opening regions **172** with a low dielectric constant material. Specifically, it is possible to fill each of the side surface opening regions **172** with a low dielectric constant material by depositing the low dielectric constant material to fill the side surface opening **172H** after the formation of the gate insulating film **171** and removing the low dielectric constant material deposited on the gate opening **170H** with anisotropic etching or the like.

MODIFICATION EXAMPLE

(81) It is also possible to form the low-Ns region **131** in a method other than the method described with reference to FIG. 5E. A modification example of the method of forming the low-Ns region **131** is described with reference to FIGS. 6, 7A, and 7B. FIG. 6 is a vertical cross-sectional view describing the modification example of the method of forming the low-Ns region **131**. FIG. 7A is an energy band diagram of the semiconductor device **100** in which fixed electric charge has not yet been applied to the barrier layer **140**. FIG. 7B is an energy band diagram of the semiconductor device **100** in which the fixed electric charge has already been applied to the barrier layer **140**. It is to be noted that, in FIGS. 7A and 7B, E_c represents the energy level at the lower end of the conduction band, E_v represents the energy level at the upper end of the valence band, and E_f represents the Fermi level.

(82) As illustrated in FIG. 6, it is possible to form the low-Ns region **131** in the channel layer **130** by performing oxidization treatment (i.e., ashing) on the barrier layer **140** exposed by the gate opening **170H** and the side surface opening **172H** with N.sub.2 plasma or O.sub.2 plasma and

generating fixed electric charge **142** in the barrier layer **140**.

(83) Specifically, as indicated by a comparison between FIGS. 7A and 7B, the fixed electric charge **142** of the barrier layer **140** pushes up the energy band of the channel layer **130** and this decreases the carriers accumulated in the channel layer **130** as the two-dimensional electron gas layer (2DEG). This makes it possible to form the low-Ns region **131** having reduced carrier density in the channel layer **130** corresponding to the barrier layer **140** having the fixed electric charge **142** added thereto. It is therefore possible to form the low-Ns region **131** in the channel layer **130** corresponding to the barrier layer **140** exposed by the gate opening **170H** and the side surface opening **172H**.

(84) In the method described above, it is possible to form the low-Ns region **131** with no mask or the like added because the low-Ns region **131** is formed in the channel layer **130** by using the gate opening **170H** and the side surface opening **172H**. It is to be noted that an end of the low-Ns region **131** may align with an end of the side surface opening **172H**. Alternatively, an end of the low-Ns region **131** may extend outward from an end of the side surface opening **172H**.

2. SECOND EMBODIMENT

(85) Subsequently, a semiconductor module according to a second embodiment of the present disclosure is described with reference to FIG. 8. FIG. 8 is a schematic perspective view of a configuration of a semiconductor module **1** according to the second embodiment of the present disclosure.

(86) As illustrated in FIG. 8, the semiconductor module **1** according to the present embodiment is an antenna-integrated module in which, for example, array-shaped edge antennas **20** and front-end components are implemented as one module. The front-end components include a switch **10**, a low-noise amplifier **41**, a bandpass filter **42**, a power amplifier **43**, and the like. The semiconductor module **1** may be used, for example, as a transceiver for wireless communication.

(87) The semiconductor module **1** includes, for example, the semiconductor device **100** according to the first embodiment as a transistor included in the switch **10**, the low-noise amplifier **41**, the power amplifier **43**, or the like. For example, the fifth-generation mobile communication (5G), which uses radio waves in a higher frequency band, has a greater propagation loss of radio waves. The semiconductor module **1** supporting 5G is thus requested to transmit radio waves with higher power. The semiconductor module **1** including the semiconductor device **100** according to the first embodiment is allowed to have higher withstand voltage performance. It is therefore possible to perform wireless communication with high power, low power consumption, and high reliability. In other words, it is possible to use the semiconductor module **1** more favorably for the fifth-generation mobile communication (5G).

3. THIRD EMBODIMENT

(88) Next, a wireless communication apparatus according to a third embodiment of the present disclosure is described with reference to FIG. 9. FIG. 9 is a block diagram illustrating a configuration of a wireless communication apparatus **2** according to the third embodiment of the present disclosure.

(89) As illustrated in FIG. 9, the wireless communication apparatus **2** includes an antenna ANT, an antenna switch circuit **3**, a high-power amplifier HPA, a radio frequency integrated circuit RFIC (Radio Frequency Integrated Circuit), a base band section BB, a sound output section MIC, a data output section DT, and an interface section I/F (e.g., wireless LAN (Wireless Local Area Network: W-LAN), Bluetooth (registered trademark), or the like). The wireless communication apparatus **2** is, for example, a mobile phone system having multiple functions including voice, data communication, LAN connection, and the like.

(90) Upon transmission, the base band section BB in the wireless communication apparatus **2** outputs a transmission signal to the antenna ANT through the radio frequency integrated circuit RFIC, the high-power amplifier HPA, and the antenna switch circuit **3**. In addition, upon reception, the antenna ANT in the wireless communication apparatus **2** inputs a reception signal to the base

band section BB through the antenna switch circuit **3** and the radio frequency integrated circuit RFIC. The reception signal processed by the base band section BB is outputted, for example, from the sound output section MIC, the data output section DT, or the interface section I/F to the outside of the wireless communication apparatus **2**.

(91) The wireless communication apparatus **2** includes the semiconductor device **100** according to the first embodiment as a transistor included in the antenna switch circuit **3**, the high-power amplifier HPA, the radio frequency integrated circuit RFIC, the base band section BB, or the like. This allows the wireless communication apparatus **2** to have higher withstand voltage performance. It is therefore possible to perform wireless communication with high power, low power consumption, and high reliability.

(92) The technology according to the present disclosure has been described above with reference to the embodiments. However, the technology according to the present disclosure is not limited to the embodiments or the like described above. A variety of modifications are possible.

(93) For example, in the embodiments described above, the example has been described in which the semiconductor device **100** includes a GaN-based compound semiconductor, but the technology according to the present disclosure is not limited to the example described above. For example, the semiconductor device **100** may include a GaAs-based compound semiconductor.

(94) Further, not all of the components and operations described in the embodiments are necessary as the components and operations according to the present disclosure. For example, among the elements in the respective embodiments, elements not recited in an independent claim based on the most generic concept of the present disclosure are to be understood as optional components.

(95) The terms used throughout this specification and the appended claims should be construed as “non-limiting” terms. For example, the term “including” or “included” should be construed as “not limited to what is described as being included”. The term “having” should be construed as “not limited to what is described as having”.

(96) The terms used in this specification are used merely for the convenience of description and include terms that are not used to limit the configuration and the operation. For example, the terms such as “right”, “left”, “up”, and “down” only indicate directions in the diagrams being referred to. In addition, the terms “inside” and “outside” only indicate a direction toward the center of a component of interest and a direction away from the center of a component of interest, respectively. The same applies to terms similar to these and terms with the similar purpose.

(97) It is to be noted that the technology according to the present disclosure may have configurations as follows. The technology according to the present disclosure having the following configurations makes it possible to form the low-Ns region having reduced carrier density in the channel layer corresponding to the barrier layer exposed by the gate opening and the side surface opening regions. This allows the semiconductor device to reduce an electric field generated between the gate electrode and the channel layer by using the low-Ns region. It is therefore possible to further increase the gate withstand voltage. Effects attained by the technology according to the present disclosure are not necessarily limited to the effects described herein, but may include any of the effects described in the present disclosure.

(98) (1)

(99) A semiconductor device including: a channel layer including a first nitride semiconductor; a barrier layer including a second nitride semiconductor, the barrier layer being provided on the channel layer; a source electrode and a drain electrode that are provided above the barrier layer; a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode; a side surface opening region that is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode; and a low-Ns region that is provided in the channel layer in correspondence with a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.

- (2)
(100) The semiconductor device according to (1), in which the gate electrode is provided above the barrier layer with a gate insulating film interposed in between.
- (101) (3)
(102) The semiconductor device according to (1) or (2), further including an insulating layer that fills a region between the gate electrode and the source electrode or the drain electrode, in which the side surface opening region is provided as a side surface opening of the insulating layer.
- (103) (4)
(104) The semiconductor device according to any one of (1) to (3), in which the low-Ns region is provided in the channel layer immediately below the gate electrode and the side surface opening region.
- (105) (5)
(106) The semiconductor device according to any one of (1) to (4), in which the carrier density of the low-Ns region is 50% or more and 90% or less of the carrier density of the other region of the channel layer.
- (107) (6)
(108) The semiconductor device according to any one of (1) to (5), in which the side surface opening region is at least provided on the side surface of the gate electrode between the drain electrode and the gate electrode.
- (109) (7)
(110) The semiconductor device according to (6), in which the respective side surface opening regions are provided on both of the side surfaces of the gate electrode between the source electrode and the drain electrode, and the gate electrode.
- (111) (8)
(112) The semiconductor device according to (7), in which a size of the side surface opening region provided between the drain electrode and the gate electrode is larger than a size of the side surface opening region provided between the source electrode and the gate electrode.
- (113) (9)
(114) The semiconductor device according to any one of (1) to (8), in which the side surface opening region is provided over an entire circumference of the side surfaces of the gate electrode.
- (115) (10)
(116) The semiconductor device according to any one of (1) to (9), in which the side surface opening region is provided to have a width of 100 nm or less from the gate electrode.
- (117) (11)
(118) The semiconductor device according to any one of (1) to (10), in which the barrier layer includes AlInN as the second nitride semiconductor.
- (119) (12)
(120) A semiconductor module including a semiconductor device, in which the semiconductor device includes a channel layer including a first nitride semiconductor, a barrier layer including a second nitride semiconductor, the barrier layer being provided on the channel layer, a source electrode and a drain electrode that are provided above the barrier layer, a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode, a side surface opening region that is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode, and a low-Ns region that is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.
- (13)
(121) A wireless communication apparatus including a semiconductor device, in which the semiconductor device includes a channel layer including a first nitride semiconductor, a barrier

layer including a second nitride semiconductor, the barrier layer being provided on the channel layer, a source electrode and a drain electrode that are provided above the barrier layer, a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode, a side surface opening region that is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode, and a low-Ns region that is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.

(122) This application claims the priority on the basis of Japanese Patent Application No. 2020-123446 filed with Japan Patent Office on Jul. 20, 2020, the entire contents of which are incorporated in this application by reference.

(123) It should be understood by those skilled in the art that various modifications, combinations, sub-combinations, and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

Claims

1. A semiconductor device comprising: a channel layer including a first nitride semiconductor; a barrier layer including a second nitride semiconductor, the barrier layer being provided on the channel layer; a source electrode and a drain electrode that are provided above the barrier layer; a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode; a side surface opening region that is provided on at least one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode; and a low-Ns region that is provided in the channel layer in correspondence with a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.
2. The semiconductor device according to claim 1, wherein the gate electrode is provided above the barrier layer with a gate insulating film interposed in between.
3. The semiconductor device according to claim 1, further comprising an insulating layer that fills a region between the gate electrode and the source electrode or the drain electrode, wherein the side surface opening region is provided as a side surface opening of the insulating layer.
4. The semiconductor device according to claim 1, wherein the low-Ns region is provided in the channel layer immediately below the gate electrode and the side surface opening region.
5. The semiconductor device according to claim 1, wherein the carrier density of the low-Ns region is 50% or more and 90% or less of the carrier density of the other region of the channel layer.
6. The semiconductor device according to claim 1, wherein the side surface opening region is at least provided on the side surface of the gate electrode between the drain electrode and the gate electrode.
7. The semiconductor device according to claim 6, wherein the respective side surface opening regions are provided on both of the side surfaces of the gate electrode between the source electrode and the drain electrode, and the gate electrode.
8. The semiconductor device according to claim 7, wherein a size of the side surface opening region provided between the drain electrode and the gate electrode is larger than a size of the side surface opening region provided between the source electrode and the gate electrode.
9. The semiconductor device according to claim 1, wherein the side surface opening region is provided over an entire circumference of the side surfaces of the gate electrode.
10. The semiconductor device according to claim 1, wherein the side surface opening region is provided to have a width of 100 nm or less from the gate electrode.
11. The semiconductor device according to claim 1, wherein the barrier layer includes AlInN as the second nitride semiconductor.

12. A semiconductor module comprising a semiconductor device, wherein the semiconductor device includes a channel layer including a first nitride semiconductor, a barrier layer including a second nitride semiconductor, the barrier layer being provided on the channel layer, a source electrode and a drain electrode that are provided above the barrier layer, a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode, a side surface opening region that is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode, and a low-Ns region that is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.

13. A wireless communication apparatus comprising a semiconductor device, wherein the semiconductor device includes a channel layer including a first nitride semiconductor, a barrier layer including a second nitride semiconductor, the barrier layer being provided on the channel layer, a source electrode and a drain electrode that are provided above the barrier layer, a gate electrode that is provided above the barrier layer between the source electrode and the drain electrode, a side surface opening region that is at least provided on one of side surfaces of the gate electrode between the source electrode or the drain electrode and the gate electrode, and a low-Ns region that is at least provided in the channel layer corresponding to a planar region provided with the gate electrode and the side surface opening region, the low-Ns region having lower carrier density than carrier density of another region of the channel layer.
